

CI/CD Deployment Assignment

This document provides a complete sample solution for deploying Flask and Express on an EC2 instance and automating deployment with Jenkins.

Part 1: Launch EC2, install Python, Node.js, Git, clone Flask and Express repositories, install dependencies, run Flask on port 5000 and Express on port 3000 using PM2/systemd.

Part 2: Install Jenkins, Git, NodeJS and Python plugins, create separate Jenkins pipelines to pull code, install dependencies, restart apps, and configure GitHub webhooks.

Sample Flask Jenkinsfile:

```
pipeline { agent any stages { stage('Checkout'){steps{git 'FLASK_REPO'}} stage('Install'){steps{sh 'pip install -r requirements.txt'}} stage('Deploy'){steps{sh 'pm2 restart flask-app || pm2 start app.py --name flask-app'}}}}
```

Sample Express Jenkinsfile:

```
pipeline { agent any stages { stage('Checkout'){steps{git 'EXPRESS_REPO'}} stage('Install'){steps{sh 'npm install'}} stage('Deploy'){steps{sh 'pm2 restart express-app || pm2 start npm --name express-app -- start'}}}}
```

Deliverables: Running EC2 instance, architecture diagram, Jenkinsfiles, README, screenshots of Jenkins builds and deployed applications.